THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 8: **Heredity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A plant of genotype TtRr produces gametes TR, Tr, tR and tr in equal numbers. The probability of a gamete tr is 1

(a) 1/2 **(b)** 1/8 **(c)** 1 **(d)** 1/4

2. A woman is XX and her husband is XY; their son has a disease on his X. From whom did he get the X? 1

(a) Father **(b)** Mother **(c)** Both **(d)** Neither

3. Assertion (A): All the gametes of a heterozygous plant are the same. Reason (R): A heterozygous plant carries two different alleles. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. For a dihybrid cross, the number of genotypes in the F2 is 1

(a) 16 **(b)** 8 **(c)** 9 **(d)** 4

5. Two parents have free earlobes. Their first child has attached earlobes. For the second child, the probability of attached earlobes is 1

(a) 1/2 **(b)** 3/4 **(c)** 0 **(d)** 1/4

6. Assertion (A): Mendel obtained a 9 : 3 : 3 : 1 ratio in the dihybrid cross. Reason (R): Genes for the two traits are inherited independently. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. If the F2 of a dihybrid cross has 80 wrinkled green seeds, the total number of seeds is approximately 1

(a) 720 **(b)** 1280 **(c)** 960 **(d)** 80

8. Mendel's work was not recognised for many years chiefly because 1

(a) he used mice
(b) it was in a foreign language
(c) it was ahead of the knowledge of chromosomes and genes
(d) the traits were not clear

9. Assertion (A): A woman is responsible for the birth of a girl child. Reason (R): Women have two X chromosomes. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. A true-breeding tall plant is crossed with a dwarf plant. The F2 has 1200 plants. The number of heterozygous tall plants is 1

(a) 300 **(b)** 1200 **(c)** 900 **(d)** 600

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A dihybrid cross gives 9 : 3 : 3 : 1. What fraction of F2 seeds is different from the two parental types? 2

12. Out of 640 F2 seeds in a dihybrid cross, how many are expected to be wrinkled yellow? 2

13. A black-coated animal (B dominant) and a brown-coated (bb) mate and produce all black offspring. What are the genotypes of the parents? 2

14. Two daughters of the same parents both receive the father's X chromosome but may differ in other traits. Why? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A cross between two plants gives tall and dwarf offspring in a 3 : 1 ratio. Identify the genotypes of the parents and the offspring. 3

16. In pea plants, a gene produces an enzyme that makes a growth hormone. A mutation reduces the efficiency of the enzyme. Predict the phenotype and the effect of giving the hormone externally. 3

17. Compare dominant and recessive alleles in terms of their expression in homozygous and heterozygous states. 3

OR

The F1 of a dihybrid cross RrYy is test-crossed. Give the genotypes and phenotypes of the offspring. 3

18. Describe how Mendel's use of statistical counting helped him discover the laws of inheritance. 3

OR

If all F1 plants in a cross of two plants are tall, what can be concluded about the parents? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Analyse the pedigree: both parents have free earlobes; of their four children, one has attached earlobes. Deduce the genotypes of the parents and children and the probability that the next child has free earlobes. 5

OR

If a human sperm contains 22 autosomes and one sex chromosome, and an egg has 22 autosomes and X, find the possible chromosome composition of the zygotes and the sex of the child. State why the child is a mixture of both parents. 5

20. A woman is told that she will have a boy as she has two daughters already. Evaluate the statement with respect to chromosomes, and also comment on the contribution of father and mother to the sex of the child. 5

OR

A student crosses RrYy with rryy and gets 52 round yellow, 48 round green, 51 wrinkled yellow and 49 wrinkled green. Give the ratio and conclude what it shows about the parental genotype and the gene inheritance. 5

END OF QUESTION PAPER **Total: 40 Marks**